Prabhavit Gupta



AI Data Pipeline Engineer: Python, Linux Systems & Automated ETL

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Summary

AI Data Pipeline and Systems Engineer with 3 years of production experience architecting, executing, and troubleshooting high-volume data ingestion pipelines, headless Linux server operations, and AI automation microservices. Deep expertise in writing resilient Python scripts (FastAPI, asyncio, Pydantic, requests), monitoring long-running background extraction jobs, and resolving complex runtime exceptions, connection timeouts, and JSON serialization bugs. Highly proficient managing headless Linux environments via SSH, configuring process daemons, virtual environments, and automated health checks. Dedicated, proactive engineer available immediately for full-time remote engineering roles.

Technical Expertise

- **Python & Data Pipelines:** Python (asyncio, requests, httpx, Pydantic, pandas), long-running extraction scripts, automated ETL/ELT workflows, webhooks, JSON/JSONL processing, data validation, and text normalization.
- **Linux Server Administration:** Headless Ubuntu/Debian server management, SSH key management, systemd services, tmux/nohup process monitoring, bash scripting, cron jobs, file permissions, and environment isolation.
- **API Integration & Resilience:** RESTful APIs, rate-limiting, exponential backoff retries, connection timeout recovery, token authentication, and error telemetry capture.
- **Databases & Messaging:** PostgreSQL (queries, indexing, schema migrations), Redis (task queues, key-value caching, state tracking), SQLite.
- **AI Tooling & Automation:** LLM API integrations, local model execution workflows, Model Context Protocol (MCP), automated regression testing with pytest and Playwright.
- **Languages & Availability:** English (C1 Professional, fluent), Hindi (Native), Italian (Conversational). Available immediately without notice period.

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
AI Pipeline & Systems Engineer *London, UK (remote)*

- Maintained and monitored mission-critical asynchronous Python data pipelines ingesting unstructured records from 15+ external API endpoints with 99.8% reliability.
- Debugged complex runtime exceptions, memory leaks, and third-party rate-limit throttling in production, writing automated error recovery routines.
- Administered headless cloud Linux virtual machines, managing Python virtual environments, systemd background worker daemons, and automated log rotation.
- Built Redis-backed job queues and worker pools to distribute heavy document parsing and LLM inference tasks smoothly across CPU cores.
- Designed automated health-check monitors and Slack alerting hooks notifying the team immediately upon pipeline degradation or schema mismatches.

Urlink **Apr 2025 - Jul 2025**
Data Platform Developer (Freelance) *Barcelona, Spain (remote)*

- Developed automated data collection and validation pipelines in Python, extracting and sanitizing web and API data into relational databases.
- Authored bash automation scripts and scheduled cron jobs running on headless remote Linux servers for daily data extraction and backup synchronization.
- Implemented end-to-end integration test suites with pytest to verify JSON schema consistency across multi-stage transformation steps.

Independent Technical Consultant **Aug 2023 - Feb 2024**
Python Automation Developer *Venice, Italy*

- Engineered custom data scraping and ETL microservices in Python for European SMB clients, automating manual reporting workflows.
- Deployed and maintained self-hosted database and API microservices on remote Linux VPS instances using Docker

and systemd.
- Authored technical troubleshooting guides and runbooks for client operational staff.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, statistical inference, quantitative economics, and complex systems modeling.